Limitations and claim boundaries

The boundaries here define the contribution. The central goal is to show — concretely and testably — that the categorical generalized-variational construction inspired by FedGVI (Mildner et al. 2025) has standard-Bayes client limits and a project-local log-linear-pool server identity.

Limitations and claim... (part 2)

Under the explicit bridge in sec. 8.8, the latter specializes Friston et al.'s Eq. 7 message-combination term (Friston et al. 2024), not the complete source protocol; behavior away from those limits is evaluated only under declared simulation conditions.

Limitations and claim... (part 3)

The exact identities are formal; performance and deployment claims remain conditional. Items beyond that boundary are named as future work.

Three robustness axes: theorem, heuristic, and objective

The authoritative definitions and claim owners are in sec. 6.3; this section records only what still cannot transfer. A source-conditional client theorem does not certify either server rule,

Three robustness axes (part 2)

a server recovery identity does not supply an objective or estimator-level influence bound, and objective descent plus raw-weight control does not imply truth recovery or peak-accuracy dominance. Likewise,

Three robustness axes (part 3)

conditional accuracy contrasts cannot confer a theorem on the heuristic. The evidence map in fig. 9 makes that no-transfer boundary and its nested replication units explicit.

Three robustness axes (part 4)

The remaining methodological problem is a server objective that combines competitive conditional performance with an appropriate robustness guarantee. Related robust-federation work motivates the target (Blanchard et al. 2017; Li et al. 2022) but does not certify either project server rule.

Three robustness axes (part 5)

Any such result must also survive beyond the present categorical model, declared attack grid, and single-host execution setting; future work and extended methods remain separate (sec. 11.17, sec. 16).

Scope boundaries that the evidence does not cross

The bridge is a recasting, not an upstream claim. Friston et al. (Friston et al. 2024) supply a belief-sharing operator for agents with a shared world model; Mildner et al.

Scope boundaries that the... (part 2)

(Mildner et al. 2025) supply robust federated generalized variational inference. Neither paper claims the other.

Scope boundaries that the... (part 3)

The contribution here is to recast the former as the tested KL/NLL zero-robustness recovery corner of the latter-inspired construction, then measure changes under robust losses and server reweighting.

Scope boundaries that the... (part 4)

Historical sources are conceptual, not formal support. The pre-modern sources added in sec. 11.9 support a genealogy of expectation, inverse inference, utility, and collective judgment (Huygens 1657; Bayes 1763; Laplace 1774; Bernoulli 1738; Condorcet 1785). They do not support claims about KL, NLL,

Scope boundaries that the... (part 5)

product-of-experts training, FedGVI, or `robust_aggregate`.

Those claims rest only on the modern cited formalism and the tests reported in sec. 10.1.

Scope boundaries that the... (part 6)

Single-machine federation. Federation is validated with queue transport, a single-machine OS-process helper, and loopback TCP: agents serialize beliefs, the server aggregates, and consensus is broadcast back without changing the mathematics.

Scope boundaries that the... (part 7)

The socket path exercises frame integrity and file-backed digest-verified replay validation; an optional SQLite round-ID guard survives local process restarts.

Scope boundaries that the... (part 8)

These controls do not substitute for cross-host deployment, identity-bound mTLS, a shared multi-host replay domain, discovery, long-running worker orchestration, or fault tolerance; those steps are scoped as future work (sec. 11.17).

Scope boundaries that the... (part 9)

Source-scale posterior FedGVI classification remains open. The displayed NumPy logistic-regression baseline is an exploratory point-estimate comparison, not the source paper's posterior-uncertainty BNN experiment (Mildner et al. 2025).

Scope boundaries that the… (part 10)

Its operating point was selected within the evaluated synthetic sweep to expose a mid-contamination margin (fig. 21).

Scope boundaries that the... (part 11)

The curves lose reliable ordering at the highest rate. The source theorem, not this selected finite sweep, carries the source-conditional robustness claim.

Scope boundaries that the... (part 12)

Classification baselines are point estimates, not full posteriors. The NumPy logistic-regression baseline and the PyTorch MLP complement both use point-estimate weights; no posterior covariance is computed.

Scope boundaries that the... (part 13)

A genuine mean-field variational family over the weights—diagonal-Gaussian $q(w)$ with a closed-form KL and a Monte-Carlo ELBO—is implemented by the tested `VariationalMLP` primitive in the `bnn_variational_torch` module. It recovers the point-estimate net exactly as its posterior variance vanishes.

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Scope boundaries that the... (part 14)

The source-comparable training and evaluation lane remains unexecuted: leakage-free calibration, source-dataset parity, posterior-family comparisons, and source-scale compute are still open.

Scope boundaries that the... (part 15)

Hierarchical depth does not by itself improve the base task. The two- and three-level stacks (sec. 17.9, sec. 17.11) run alternating L1/L2(/L3) inference end-to-end and resolve their added context latents above chance. On the shared location task they do not beat the flat baseline:

Scope boundaries that the… (part 16)

the paired location-accuracy gap is a small, statistically reliable negative gap at the reported seed count (fig. 31).

Scope boundaries that the... (part 17)

Depth is therefore validated as executable and consensus-preserving, not as an accuracy improvement; whether a richer policy-and-horizon task family rewards hierarchy is future work (sec. 11.20).

Scope boundaries that the... (part 18)

Discrete POMDP only—continuous or hybrid state spaces are not addressed. This work is discrete-categorical only, matching the community's worked POMDP example (Da Costa et al. 2020; Friston et al. 2024).

Scope boundaries that the… (part 19)

The limit-as-proof contract is validated off the categorical case only for a one-dimensional Gaussian-mean conjugate slice; continuous-state active inference and Gaussian belief-sharing colonies remain untested.

Scope boundaries that the... (part 20)

Temperature and divergence calibration are fixed, not learned. Coarsened posterior and safe-Bayes theory make clear that the learning-rate/temperature is part of the inference rule under misspecification (Miller and Dunson 2018; Grünwald 2012; Kleijn and Vaart 2012).

Scope boundaries that the… (part 21)

This manuscript validates the configured losses and divergences, but it does not learn an optimal coarsening radius, temperature, or divergence schedule across agents.

Scope boundaries that the... (part 22)

Real multi-machine federation, networking, and privacy cryptography. Federation here is *mathematical* (factor aggregation), not infrastructural— unlike communication-efficient or Byzantine-robust federated learning (McMahan et al. 2017; Blanchard et al. 2017).

Scope boundaries that the… (part 23)

New linguistic theory. The language-acquisition study (sec. 10.3) reproduces the Dirichlet count mechanism (eq. 18) mechanically; it does not extend the linguistics.

What the statistics can and cannot claim

The verdict is a paired comparison at a single high contamination rate (sec. 10.5.2, tbl. 7). The design concentrates power where the effect is largest.

What the statistics can and... (part 2)

It therefore identifies *where* robustness pays off, but it does not characterize contamination as a continuous function.

What the statistics can and... (part 3)

The per-rate effect and inference projections (tbls. 5, 6) describe the remaining sweep.

What the statistics can and... (part 4)

Their inferential claims do not exceed the declared BH-FDR control (Benjamini and Hochberg 1995).

What the statistics can and... (part 5)

BH-FDR controls expected false discovery proportion within the declared family, not the chance of any false positive. The matched-pairs Wilcoxon tests are rank tests under paired-difference assumptions (Wilcoxon 1945; Fay and Proschan 2010), not assumption-free proofs about raw means.

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What the statistics can and... (part 6)

Confidence intervals are percentile bootstrap (Efron and Tibshirani 1993), not analytic, and inherit that method's small-sample caveats at the lowest trial counts.

What the statistics can and... (part 7)

The power values are observed-effect design approximations useful for confirmatory sample-size planning; they are not independent evidence for the verdict and do not cover model-specification uncertainty outside the seeded simulation harness (Wasserstein and Lazar 2016).